

pythonbasics

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Python starts now

So amra python basics porbo first e
numpy testing:

```
[ ]: import numpy as np
A = np.array([[1, 2, 3],
              [4, 5, 6],
              [7, 8, 9]])
b = np.array([10, 20, 30])
C = A + b
D = A * C
column_means = np.mean(D, axis=0)

print("Matrix A:\n", A)
print("Vector b:\n", b)
print("Broadcasted Addition (A + b):\n", C)
print("Element-wise Multiplication (A * C):\n", D)
print("Column-wise Means:\n", column_means)
```

Matrix A:

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

Vector b:

```
[10 20 30]
```

Broadcasted Addition (A + b):

```
[[11 22 33]
 [14 25 36]
 [17 28 39]]
```

Element-wise Multiplication (A * C):

```
[[ 11  44  99]
 [ 56 125 216]
 [119 224 351]]
```

Column-wise Means:

```
[ 62. 131. 222.]
```

some noob python code:

```
[15]: character_name="Saklain"
character_age=21
isMale=True #capital diye shuru true
print("Hello from "+character_name+" .Ki obostha shobar??")
print("His age is: "+str(character_age)) # ekhane just character_age dile hobe
    ↳na....noob python....number ke string banate pare na nije....amader kore
    ↳nite hobe.. str() eta diye
print(character_age) #but just eta print korte chaile string kora lagbe na
character_name="Abbi" #ebhabei modify kore yaaayyy
#java types er pray shob
phrase="Adventures Of Tintin"
print(phrase.upper())
print(phrase.lower())
print(phrase.isupper())#upper kina check kore shob
print(len(phrase))
print(phrase[3])
print(phrase.index("f"))#jeta khujtesi oita na thakle error dibe
print(phrase.index("Tintin"))# emon poora string dile mainly ora oi string er
    ↳first character er index dey # exactly milte hobe with case tahole pabe
    ↳nahoy to error
print(phrase.replace("Tintin","Saklain"))#ekta new string return kkore aager ta
    ↳change hoyna
print(phrase)#tar mane ei je replace korlam eta real string ke edit kore na
```

```
Hello from Saklain .Ki obostha shobar??
His age is: 21
21
ADVENTURES OF TINTIN
adventures of tintin
False
20
e
12
14
Adventures Of Saklain
Adventures Of Tintin
```

```
[ ]: print(3*4.5)
print(10%3)
my_num=-5
print(my_num)
print(abs(my_num))
print(pow(3,2))
print(pow(4,6))
print(max(4,6))
print(min(4,6))
print(round(3.7))
```

```
print(round(3.5))#upper e chole jay
```

13.5

1

-5

5

9

4096

6

4

4

4

```
[22]: from math import *  
my_num=-5  
print(floor(3.7))  
print(ceil(3.2))  
print(sqrt(3))
```

3

4

1.7320508075688772

get input from user:

```
[25]: name=input("Enter your name")  
age=input("Enter your age: ")  
print("Hello "+name+"! You're "+age)
```

Hello Saklain! You're 21

```
[ ]: num1=input("Enter a number: ")  
num2=input("Enter another number: ")  
result=int(num1)+int(num2) #int e na nile era string concat kore dey lol #abar  
    ↪etay problem ache float input dile mara kheyee error dibe codde....so amader  
    ↪uchit float e niye jawa shob kichu ke  
print(result)
```

```
[26]: print("i have {nothing}")
```

i have {nothing}

Lists

```
[35]: siblings=["Saklain", "Abbi", "Tuba"]  
      #         0         1         2  
      #        -3        -2        -1  
print(siblings)  
print(siblings[1])
```

```
['Saklain', 'Abbi', 'Tuba']
Abbi
Tuba
Saklain
['Abbi', 'Tuba']
['oop', 'math', 'chem']
['Dsa', 'oop', 'linear algebra', 'chem', 'mecha']
```

```
['Saklain', 'Ussop', 'akif', 'mithun', 'mufeed', 'akib', 'akib', 'shuvo']
['Saklain', 'Ussop', 'akib', 'akib', 'akif', 'mithun', 'mufeed', 'shuvo']
[1, 2, 2, 2, 3, 23, 43, 45, 234, 234, 234, 3234]
['Saklain', 'Ussop', 'akib', 'akib', 'akif', 'mithun', 'mufeed', 'shuvo', 1, 2,
2, 2, 3, 23, 43, 45, 234, 234, 234, 3234]
5
2
```

```
['Saklain', 'Ussop', 'akib', 'akib', 'akif', 'mithun', 'mufeed', 'shuvo', 1, 2,
2, 3, 23, 43, 45, 234, 234, 234, 3234]
['Saklain', 'Ussop', 'akib', 'akib', 'akif', 'mithun', 'mufeed', 'shuvo', 1, 2,
2, 3, 23, 43, 45, 234, 234, 234, 3234]
['Saklain', 'Ussop', 'akib', 'akib', 'akif', 'mithun', 'mufeed', 'shuvo', 1, 2,
2, 3, 23, 43, 45, 234, 234, 234]
[]
['Saklain', 'Ussop', 'akib', 'akib', 'akif', 'mithun', 'mufeed', 'shuvo', 1, 2,
2, 3, 23, 43, 45, 234, 234, 234]
```

Tuples

```
[50]: coordinates = (4,5) # tuples immutable hoy
      #coordinates[1]=6 # change kora jayna
      print(coordinates[1])
```

5

```
[51]: coordinatelist=[(4,5), (7,8), (9,10)]
      print(coordinatelist)
```

[(4, 5), (7, 8), (9, 10)]

Functions

```
[54]: def say_hi():
      print("hello from function")

      print("Top")
      #now just call it
      say_hi()
      print("bottom")
```

Top
hello from function
bottom

```
[56]: def say_hi(name,age):
      print("Hello "+name+" , you're "+age)

      say_hi("Saklain", "21")
```

Hello Saklain , you're 21

using return keyword:

```
[ ]: def cube(num):
      return num*num*num
      print("hehe") # eta to return er pore

      print(cube(2))
```

```
result = cube(4)
print(result)
```

8
64

If statements

```
[64]: is_male=True
      is_tall=False
      if is_male and is_tall:
          print("Hey boy")
      elif is_male and not(is_tall):
          print("Shorty")
      elif not is_male:
          print("tintin")
      else:
          print("Who are you??")
```

Shorty

```
[ ]: def max_num(num1,num2,num3):
      if num1>=num2 and num1>=num3:
          return num1
      elif num2>=num3:
          return num2 # != etao ache btw
      else:
          return num3

      print(max_num(100,7,9))
```

100

Key-Value Pairs

```
[71]: month_conversions={
      "Jan" : "January",
      "Feb" : "February",
      "Mar" : "March",
      "Apr" : "April",
      "May" : "May",
      "Jun" : "June",
      "Jul" : "July",
      "Aug" : "August",
      "Sep" : "September",
      "Oct" : "October",
      "Nov" : "November",
      "Dec" : "December"
      }
```

```
print(month_conversions["Nov"])
print(month_conversions.get("Dec"))
print(month_conversions.get("Luv","not a valid key")) # na pele porer argument_
↳e ja pathaisi oita print korbe...r porer argument na dewa thakle none print_
↳korbe
```

November
December
not a valid key

While loop

```
[73]: i=1
      while i<10:
          print(i)
          i+=1
```

1
2
3
4
5
6
7
8
9

```
[74]: secret_word="Tintin"
      guess=""
      while guess!=secret_word:
          guess=input("Enter your guess: ")
      print("We got it!!!!")
```

We got it!!!!

For Loops

```
[75]: for letter in "Adventures":
      print(letter)
```

A
d
v
e
n
t
u
r
e
s

```
[76]: for friend in friends2:
      print(friend)
```

```
Saklain
Ussop
akib
akib
akif
mithun
mufeed
shuvo
1
2
2
3
23
43
45
234
234
234
```

```
[77]: for idx in range(10):
      print(idx)
```

```
0
1
2
3
4
5
6
7
8
9
```

```
[ ]: for idx in range(3,10):           #from 3 to 10 not including 10
      print(idx)
```

```
3
4
5
6
7
8
9
```

```
[79]: for idx in range(len(siblings)):
      print(siblings[idx])
```


Saklain
Abbi
Tuba

Exponenets

```
[ ]: print(2**3)
      print(2**5) #2^5 ber kore mainly

      def raise_to_power(num, power):
          ans=1
          for idx in range(abs(power)):
              ans*=num
          if power >= 0:
              return ans
          else:
              return 1/ans

      print(raise_to_power(2,5))
```

8
32
32

2D Lists

```
[90]: number_grid=[
        [1,2,3],
        [4,5,6],
        [7,8,9],
        [0]
      ]

      print(number_grid[0][2])

      for row in range(len(number_grid)):
          for col in range(len(number_grid[row])):
              print(number_grid[row][col],end=" ")
          print()

      for row in number_grid:
          for col in row:
              print(col,end=" ")
      print()

      for row in number_grid:
          print(row)
```

3
1 2 3

```
4 5 6
7 8 9
0
1 2 3 4 5 6 7 8 9 0
[1, 2, 3]
[4, 5, 6]
[7, 8, 9]
[0]
```

A translator:

```
[93]: def translate_word(phrase):
        translation=""
        for letter in phrase:
            if letter in "AEIOUaeiou":
                translation+="g"
            else:
                translation+=letter
        return translation

print(translate_word("To be or not to be... that is the question"))
```

Tg bg gr ngt tg bg... thgt gs thg qggstggn

Exception Handling

```
[98]: try:
        value=10/0
        number=int(input("Enter aa number"))
        print(number)
    except ZeroDivisionError as err:
        print(err)
    except ValueError:
        print("Invalid input")
```

division by zero

Reading from file

```
[105]: tintinFile=open("tintin.txt","r")
        print(tintinFile.readable())
        print(tintinFile.read())
        tintinFile.seek(0)
        print(tintinFile.readline())
        tintinFile.seek(0)
        print(tintinFile.readlines()[1])
        tintinFile.close()
        tintinFile=open("tintin.txt","a")
        tintinFile.write("\nSami - 23")
        tintinFile.close()
```

```
True
Saklain - 21
Mithun - 21
Akib - 20
Akif - 21
ussop - 59Sami - 23
Saklain - 21

Mithun - 21
```

Python e arekta file e lekha code just “import filename” diye import kora jay tahole amra just eta use korei onno file er func , variable use korte parbo...like filename.say_hi(“Saklain”) evabe.

Classes and Objects

```
[111]: class Student:
        def __init__(self,name,major,gpa,is_on_probation):
            self.name=name
            self.major=major
            self.gpa=gpa
            self.is_on_probation=is_on_probation

        def on_honor_roll(self):
            if self.gpa>=3.5:
                return True
            else:
                return False

class GoodStudent(Student):
    def can_create_something(self):
        print("He creates")

student1=Student("Saklain","Cse",4,False)
print(student1.name)
print(student1.major)
print(student1.on_honor_roll())

studentgood =GoodStudent("Mithun","cse",4,False)
studentgood.can_create_something()
```

```
Saklain
Cse
True
He creates
```